

Linker-Guided Growth of Single-Crystal Covalent Organic Frameworks

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Cite This: *J. Am. Chem. Soc.* 2024, 146, 3449–3457



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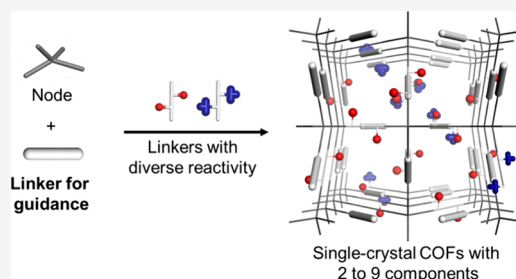


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ABSTRACT: The core features of covalent organic frameworks (COFs) are crystallinity and porosity. However, the synthesis of single-crystal COFs with monomers of diverse reactivity and adjustment of their pore structures remain challenging. Here, we show that linkers that can react with a node to form single-crystal COFs can guide other linkers that form either COFs or amorphous polymers with the node to gain single-crystal COFs with mixed components, which are homogeneous on the unit cell scale with controlled ratios. With the linker-guided crystal growth method, we created nine types of single-crystal COFs with up to nine different components, which are more complex than any known crystal. The structure of the crystal adapted approximately to that of the main component, and its pore volume could be expanded up to 8.8%. Different components lead to complex and diverse pore structures and offer the possibilities to gain positive synergies, as exemplified by a bicomponent COF with 2200 and 733% SO₂ uptake capacity of that of the two pure-component counterparts at 298 K and 0.002 bar. The selectivity for separation of SO₂/CO₂ ranges from 1230 to 4247 for flue gas based on ideal adsorbed solution theory, recording porous crystals. The bicomponent COF also exhibits a 1300% retention time of its pure-component counterparts for SO₂ in a dynamic column breakthrough experiment for deep desulfurization.



INTRODUCTION

Covalent organic frameworks (COFs) are an emerging class of organic polymers that allow atomically precise design and integration of nodes and linkers into ordered networks with permanent porosity through covalent linkages.^{1,2} They promise as model systems to understand the interaction of photons, electrons, holes, excitons, and guest molecules with pores and the structure–property relationships of organic polymers in various application scenarios.^{3–5} The promise has been greatly hindered by the lack of a method to precisely tune the pore structure of single-crystal COFs. In addition, only a few strategies could lead to single-crystal COFs and none of them could create single-crystal COFs with nodes or linkers of diverse reactivity.^{6–9} A potential way to tune the pore structure of single-crystal COFs is postsynthesis modification, which would be severely limited by having complete reactions at the linkers or nodes.¹⁰

Here, we show that linkers that can react with a node to form single-crystal COFs can guide other linkers that form either COFs or amorphous polymers with the node to gain mixed-component single-crystal COFs with bulk purity and controlled ratios (Figure 1a). 1,4-Phthalaldehyde (linker-A) was selected as the guide, since it can condense with tetrakis(4-aminophenyl)methane (node) to form COF-300 (termed COF-A), which has been one of the most studied COFs with clear crystal structures.⁷ 2,5-Dihydroxyterephthalaldehyde

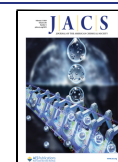
(linker-B) was selected as the first follower since it can condense with tetrakis(4-aminophenyl)methane to form COF-301 (termed COF-B) whose single-crystal structure is also clear.⁹ We showed that the bicomponent single-crystal COF-AB adapted its crystal structure to either COF-A or COF-B depending on the ratios of different components, which could be controlled (termed “a”, “b”, etc., depending on the ratios of different components). The controlled ratios and single crystallinity distinguish COF-AB from previously reported COFs with different components, which are required to be at specific ratios and partial crystallinity.^{11,12} We went further with linkers of diverse reactivity to obtain COF-C, COF-D, and COF-E and amorphous polymers (termed as aCOF) aCOF-F, aCOF-G, and aCOF-H by condensation of tetrakis(4-aminophenyl)methane with 2,5-bis(prop-2-yn-1-yloxy)-terephthalaldehyde (linker-C), 2,3,5,6-tetrafluoroterephthalaldehyde (linker-D), thieno[3,2-*b*]thiophene-2,5-dicarbaldehyde (linker-E), 2,5-dichloroterephthalaldehyde (linker-F), 2,5-dibromoterephthalaldehyde (linker-G), and 2-nitroterephthal-

Received: November 21, 2023

Revised: January 11, 2024

Accepted: January 11, 2024

Published: January 25, 2024



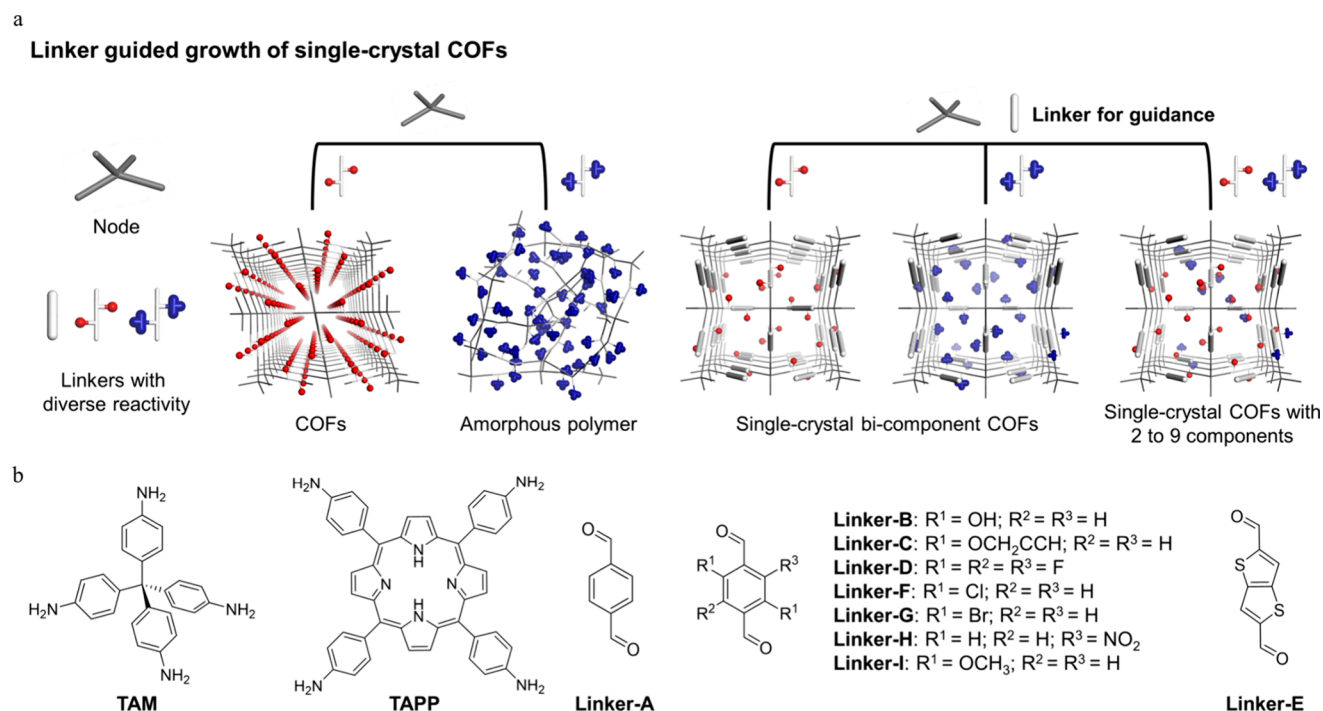


Figure 1. Linker-guided growth of single-crystal COFs. (a) Schematic illustration of linker-guided growth of single-crystal COFs with different linkers as followers, which react with the same node as the guider to form either crystalline COFs or amorphous polymers. With the developed method, single-crystal COFs with up to nine different components could be obtained. (b) Chemical structure of nodes, tetrakis(4-aminophenyl)methane (TAM), and 4,4',4'',4'''-(porphyrin-5,10,15,20-tetra-yl)tetraaniline (TAPP), linker-A-I, and single-crystal COFs with different components. Linker-A: terephthalaldehyde; linker-B: 2,5-dihydroxyterephthalaldehyde; linker-C: 2,5-bis(prop-2-yn-1-yloxy)-terephthalaldehyde; linker-D: 2,3,5,6-tetrafluoroterephthalaldehyde; linker-E: thieno[3,2-*b*]thiophene-2,5-dicarbaldehyde; linker-F: 2,5-dichloroterephthalaldehyde; linker-G: 2,5-dibromoterephthalaldehyde; linker-H: 2-nitroterephthalaldehyde; linker-I: 2,5-dimethoxyterephthalaldehyde.

aldehyde (linker-H), respectively. Linker-A could guide all these linkers to react with tetrakis(4-aminophenyl)methane to form bicomponent single-crystal COF-AC, COF-AD, COF-AE, COF-AF, COF-AG, and COF-AH (Methods and Supplementary Section 1). These bicomponent single-crystal COFs exhibit different crystal lattices from single-component COFs, which distinguish them from multivariate metal-organic frameworks and solid solutions of different compounds that have the same or very similar lattice structures as their single-component counterparts.^{13,14} We went further to use linker-A to guide the polymerization of linkers that react with the node to form either crystalline COFs or amorphous polymers to construct multicomponent single-crystal COFs with up to nine different components (Methods and Supplementary Section 1), which have more components than any known crystals.^{13,15} Different components with controlled ratios within one phase precisely adjust the pore structure and related properties of the obtained single-crystal COFs. COF-AB-c and COF-ABDI-a were then selected to show the diversity of pore structures, and COF-AB-c was selected to show the superiority of single-crystal COFs with different components using the separation and recovery of SO_2 from flue gas for illustration.

RESULTS AND DISCUSSION

We started with the construction of COF-AB-c with equivalent molar amounts of tetrakis(4-aminophenyl)methane, linker-A, and linker-B in aqueous solution using micelles of the palmitoylglycine-regulated synthesis method recently developed by us.⁹ The crystals were obtained as uniform cuboids with a typical length of approximately 6.5 μm (Figure 2a). The

reaction yield was 90%, while the maximum yield was 50% if only one linker was reacted, indicating the formation of a new material involving both linkers. The solid-state ^{13}C cross-polarization magic angle spinning nuclear magnetic resonance spectroscopy (^{13}C CP/MAS NMR) spectrum of COF-AB-c showed resonances at 65.0, 134.0, 153.0, and 158.0 ppm (ppm) (Figure 2b), characteristic of the unique carbon atoms of tetrakis(4-aminophenyl)methane, linker-A, linker-B, and imine linkage, respectively. The characteristic peaks of carbon atoms of free aldehyde groups in linker-A and linker-B at 201 and 196 ppm and those of bare amine groups in tetrakis(4-aminophenyl)methane at 129 ppm were not detected (Figure 2b). The spectrum indicated that there were no free building units and they were all covalently connected through imine bonds in COF-AB-c.

To test the homogeneity of the distribution of the different components, we performed confocal Raman spectroscopy with a laser spot size of 0.7 μm on three different areas on a COF-AB-c crystal, as well as on different crystals¹⁶ (Figure 2c). No obvious difference in either the Raman band shift or relative intensity of the spectra was observed, which indicated that COF-AB-c did not contain macroscopic domains of either COF-A or COF-B and there was no noticeable difference in their distributions among different crystals.

The high crystallinity of COF-AB-c was evident from multiple sharp powder XRD patterns. There was no overlap between the characteristic peak positions of COF-AB-c and those of COF-A and COF-B (Supplementary Figure 1a), suggesting that COF-AB-c is a single-phase material. In contrast, different small organic molecules or organic polymers usually form cocrystals or amorphous mixtures.^{15,17} All the

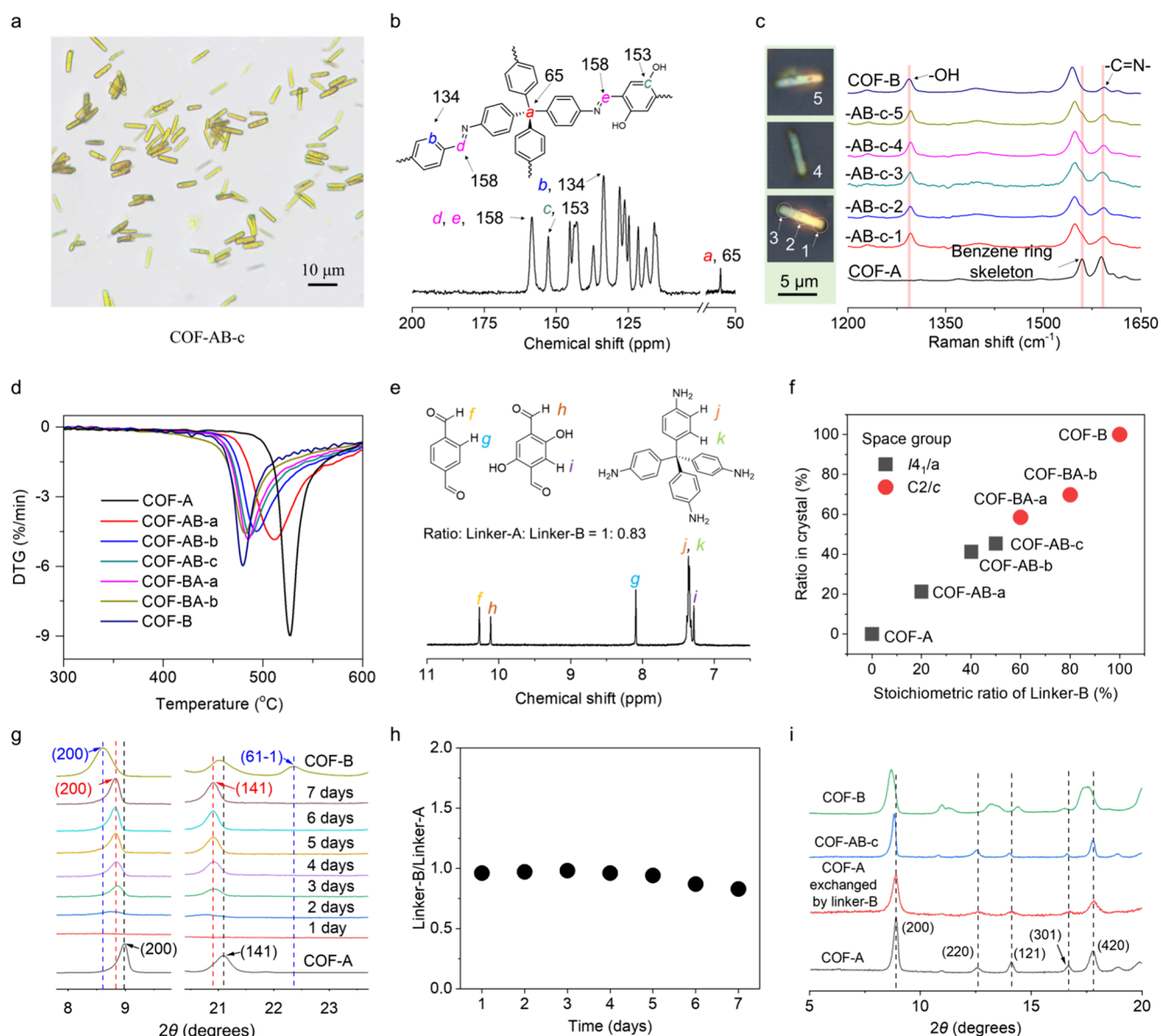


Figure 2. Chemical and phase structure of COF-AB-c. (a) Optical micrographs of COF-AB-c. (b) ^{13}C CP/MAS NMR spectrum showing the unique resonance of COF-AB-c. (c) Optical image and Raman spectra of the marked points of the COF-AB-c crystals. (d) Differential thermal gravities of COF-A, COF-AB-a, COF-AB-b, COF-AB-c, COF-BA-a, COF-BA-b, and COF-B. The linker ratios in different bicomponent COFs are presented in (f). COF-AB means it adapts to the space group of $I4_1/a$, and COF-BA means it adapts to the space group of COF-B (i.e., $C2/c$). (e) ^1H NMR spectrum of a DCl-digested solution of the COF-AB-c crystal. (f) Reaction stoichiometry and ratio of linker-B determined for COF-ABs and COF-BAs. (g, h) Powder XRD patterns (g) and ^1H NMR spectrum of a DCl-digested solution (h) of COF-AB-c with different reaction times. (i) Powder XRD patterns of COF-A, COF-A exchanged by linker-B, COF-AB-c, and COF-B.

powder XRD patterns of COF-AB-c could be indexed to the space group of $I4_1/a$, and Le Bail refinement gave refined unit cell parameters of $a = 20.04649 \text{ \AA}$ and $c = 8.91513 \text{ \AA}$ (Supplementary Figure 1b).¹⁸ We then used these crystal parameters and the chemical structure of COF-A to simulate the powder XRD patterns of COF-AB-c and found that the simulated patterns fit very well with the experimental data (Supplementary Figure 1b). Single-crystal electron diffraction was then performed to prove the single crystallinity of COF-AB-c and resolve its atomic structure (Figure 3a, Supplementary Figure 1c and Supplementary Section 2). It is a single crystal with a tetragonal space group of $I4_1/a$ ($a = 19.92 \text{ \AA}$, $c = 8.64 \text{ \AA}$) and sevenfold interpenetration. The COF has one-dimensional straight pores along the interpenetration direction

of its network. The refined data resolution was $\sim 0.9 \text{ \AA}$. These results suggest that linker-A and -B are well dispersed within COF-AB-c, with no signs of phase separation over the unit cell size.^{13,18} The absence of phase separation was further confirmed by the appearance of only one peak of decomposition rate ($\sim 490^\circ\text{C}$), which is between those of COF-A ($\sim 539^\circ\text{C}$) and COF-B ($\sim 480^\circ\text{C}$) in the differential thermogravimetric curve of COF-AB-c (Figure 2d). In addition, COF-AB-c retained the porosity and structural stability of the original COF-A structure and gained a Brunauer–Emmett–Teller surface area of $1230 \text{ m}^2 \text{ g}^{-1}$, as indicated by N_2 adsorption at 77 K (Supplementary Figure 1d).

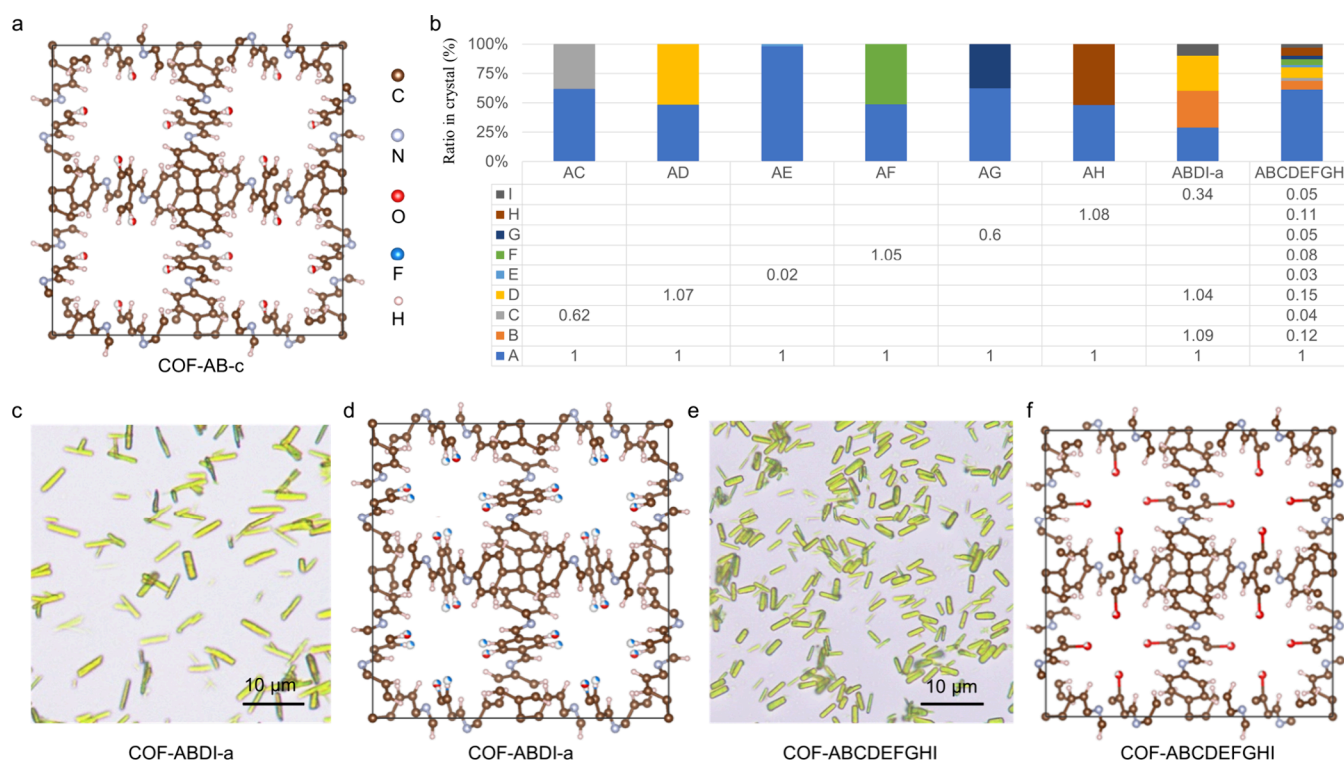


Figure 3. Single-crystal structure of COF-AB-c and COF-ABDI-a and generality of the method. (a) Schematic diagram of the refined COF-AB-c structure. (b) Ratio in crystal of COF-AC, COF-AD, COF-AE, COF-AF, COF-AG, COF-AH, COF-ABDI-a and COF-ABCDEFghi. (c–f), Optical micrographs and refined structure of COF-ABDI-a (c, d) and COF-ABCDEFghi (e, f). In all cases, the ratios were normalized to 1 for linker-A. Linker ratios are in equivalent molar amounts unless otherwise stated. COF-ABCDEFghi (A:B:C:D:E:F:G:H:I) = 1:0.25:0.25:0.25:0.25:0.25:0.25:0.25:0.25.

Since the materials are of great purity and homogeneity, the precise ratio of the two components was then obtained from the ^1H NMR spectra of a DCI-digested solution of the COF-AB-c crystal (Figure 2e). Resonances with the predicted peaks occurred at the expected chemical shifts for the protons of the unique carbons of both linkers. By integrating the intensities of the peaks, we obtained the ratio of linker-A to linker-B as 1.00:0.83 for COF-A to COF-B. Notably, repeated experiments with different weight amounts of the crystals provided comparable results, further confirming the homogeneity of the distribution of different components within COF-AB-c. We then systematically tuned the ratios of both linkers in the starting mixture and obtained COF-ABs as uniform cuboids with controlled component/linker ratios (Figure 2f, Supplementary Figure 1e and Supporting Information Table 2). No phase separation was indicated by differential thermogravimetric analysis (Figure 2d). Le Bail refinement indicated that the volume of the unit cell of the crystals expanded accordingly to accommodate the hydroxyl groups from linker-B without changing the framework topology and interpenetration degree (Supplementary Table 3 and Supplementary Section 3). Interestingly, when the molar ratio percentage of linker-B reached 58.5% and beyond, the crystal adopted the space group of COF-B (i.e., $C2/c$), indicating that linker-A was well dispersed into the structure of COF-B in these COF-BAs (Figure 2f). Note that the lattice parameters of the bicomponent (A and B) COFs as well as their gradual change from COF-A to COF-B distinguish them from multivariate metal–organic frameworks and solid solutions of different compounds that have the same or very similar lattice structures as their single-component counterparts.^{13,14}

To gain insight into the mechanism for the formation of COF-AB-c, we performed a time-sequence study (Figure 2g,h). By halting the reaction at the end of the first day, second day, third day, fourth day, fifth day, sixth day, and seventh day, we collected the resulting materials and analyzed them with powder X-ray diffraction (XRD) (Figure 2g). Pronounced Bragg peaks of (200) and (141) for COF-AB-c were observed clearly after 2 days, and their intensities increased over time, indicating a gradual crystallization process. During the whole process, no Bragg peaks of (200) and (141) for COF-A and (200) and (61-1) for COF-B occurred, indicating that the node reacted with both linker-A and linker-B concurrently leading to direct formation of single crystals. The mechanism was further proved by the ratios of linker-B to linker-A for the DCI-digested COF-AB-cs obtained at different reaction times with ^1H NMR spectra, which changed smoothly from ~ 0.98 to 0.83 (Figure 2h). In contrast, linker exchange usually resulted in significant discrepancy in the ratios of different linkers.⁶ We conducted contrast experiments using COF-A as a seed. Subsequently, linker-B was added to the solution. Neither COF-B nor COF-AB was obtained, as indicated by powder XRD, despite varying concentrations of added linker-B. Similarly, when linker-B was introduced into the solution, neither COF-B nor COF-AB was obtained, as confirmed by ^1H NMR spectra, suggesting that either no linker exchange occurred or linker exchange could not alter the lattice volume of COF-A, given that the powder XRD peak positions remained identical to those of COF-A. In addition, the pore volume of COF-AB-c was 2.4% higher than that of COF-A as indicated by the single crystal structure. The linker-guided crystallization method differs from the single-crystal trans-

formation method via linker exchange, which typically offers crystals with domains of different components and phase separation during the linker exchange process besides the difficulty to have full linker exchange.⁶

To show the generality of the linker-A-guided crystallization, linkers with different functional groups and lengths that have diverse reactivity with nodes were explored (Figure 1b, Supplementary Figures 2 and 3). Among them, linker-C, linker-D, and linker-E could be polymerized with tetrakis(4-aminophenyl)methane to form COF-C, COF-D, and COF-E, respectively, with micelles of the palmitoylglycine-regulated synthesis method. Crystalline COF-C, COF-D, and COF-E could not be obtained with the current solvothermal synthesis method.⁹ Only amorphous polymers were obtained when linker-F, linker-G, and linker-H were polymerized with tetrakis(4-aminophenyl)methane regardless of the synthesis method (Supplementary Figure 3). When linker-A guided the polymerization of each of these linkers with tetrakis(4-aminophenyl)methane, single-crystal bicomponents COF-AC, COF-AD, COF-AE, COF-AF, COF-AG, and COF-AH were obtained (Supplementary Figures 2 and 3) despite the diverse reactivity of the different linkers, the bulky size of the functional group in linker-C, and the longer length of linker-E, which significantly changed the crystallization behavior and lattice parameters of the products as well as increased their internal tension. When the added linker ratios were 1:1, the actual ratios were 0.62, 1.07, 0.02, 1.05, 0.60, and 1.08, assuming that linker-A was always 1 in COF-AC, COF-AD, COF-AE, COF-AF, COF-AG, and COF-AH, respectively (Figure 3b and Supplementary Section 1). The volume of their lattice expanded by 8.8, 4.4, 1.1, 5.2, 4.2, and 4.9%, respectively (Supplementary Table 4 and Supporting Information, Section 3). The volume expansion is significant since the tolerance limit of volume change for inorganic crystals is normally less than 5%.^{19,20} The formation of the bicomponent COFs was further confirmed by ¹³C CP/MAS NMR, and all the powder XRD patterns could be indexed to the space group of *I*4₁/*a*, indicating the formation of single-phase materials (Supplementary Table 4 and Supplementary Section 3).

Next, we used linker-A to guide the polymerization and crystallization of multiple linkers with tetrakis(4-aminophenyl)methane, and single-crystal COFs with four and nine different components were obtained (Methods and Supplementary Section 1). We addressed the morphology, chemical compositions, homogeneity of chemical composition, crystallinity, volume expansion in comparison with COF-A, porosity, pore size distribution, thermal stability, and ratios of linkers of both COFs with optical microscopy, ¹³C CP/MAS NMR, Raman spectroscopy, single-crystal electron diffraction, powder XRD, Le Bail refinement, gas absorption and desorption experiment, thermogravimetric and differential thermogravimetric analyses, and ¹H NMR spectra of DCl-digested crystals (Supplementary Sections 2–7). Both COFs were well presented, and we used COF-ABDI as a demo. COF-ABDIs are single-phase materials with four components at varied ratios, which can be controlled by tuning the reaction stoichiometry (Figure 3c,d, Supplementary Tables 5 and 6, and Supporting Information Figure 4). The molar amounts of linker-B, linker-D, and linker-I in COF-ABDIs are approximately 100–115, 94–110, and 34%–59% of the reaction stoichiometry, assuming that all linker-A was polymerized. The lower amount of linker-I in COF-ABDIs was due to the larger size of its functional groups, which could cause higher steric

interactions with others. The integration of the four linkers in COF-ABDIs led to a pore size distribution that is not the superposition of COF-A, COF-B, COF-D, and COF-I (Supplementary Figure 4a).²¹ The volume expansion ratio of the COF-ABDIs in comparison to that of COF-A ranged from 5.3 to 7.7% (Supplementary Table 6). It is noteworthy that COF-ABCDEFGHI (Figure 3e,f) has three components that cannot be crystallized by themselves, and it has more components than any known crystals.^{13,15,22}

We further synthesized a film composed of single-crystal bicomponent two-dimensional (2D) COFs, 2DCOF-AB (Supplementary Figure 5a). Raman spectra of 2DCOF-AB at 1166 and 1663 cm^{−1} showed characteristic peaks of 2DCOF-A and 2DCOF-B, respectively (Supplementary Figure 5b), indicating the formation of bicomponent 2DCOF. Transmission electron microscopy (TEM) indicated that the film of 2DCOF-AB was composed of different domains connected by grain boundaries, which exhibited bright contrast (Supplementary Figure 5c). Six different areas of a domain were selected for selective area electron diffraction (SAED). The SAED patterns showed the same orientation, indicating the single crystallinity of the domain.

Different components with controlled ratios within one phase would lead to complex and diverse pore structures and offer the possibility of discovering new properties that surpass those of their pure-component counterparts. We chose COF-AB-c and COF-ABDI-a to show the diversity of pore structures and COF-AB-c to show the superiority of single-crystal COFs with different components using the separation and recovery of SO₂ from flue gas for illustration. There are six possible basic pore structures for COF-AB-c based on its crystal structure, and they all have straight channels (Supplementary Figure 6a and Supporting Information, Table 7). When no linker-B occurs in the unit cell, the size of the pore is 5.89 Å. The pore size decreased, and the pore shape deformed gradually with the increase in the number of linker-B within one lattice, and the minimum pore size amounted to 5.39 Å when the number reached four. It is assumed that different components/linkers are arranged in a specific sequence to determine pore structures.²² In addition to the similarity, the pore structure of COF-ABDI becomes more complex (Supplementary Figure 6b and Supporting Information Table 7). If its lattice only contains linker-A, the channel is straight; if it only contains linker-B, the straight channel will be slightly distorted; if it only contains linker-D, the channel will be further distorted; and if it only contains linker-I, the channel will be helical. The controlled ratios and random combination of different linkers would lead to atomically precise adjustment of the pore shape, size, and pore wall structures for specific applications.

The emission of sulfur dioxide (SO₂) from flue gas is harmful to the environment and human health,²³ but SO₂ is also an important source for the industrial production of sulfuric acid if it is separated, purified, and recovered.²⁴ Trace SO₂ deactivates amine-based scrubbers, causes pipeline corrosion during gas transportation, and irreversibly poisons catalysts for selective NO_x reduction and CH₄ combustion.^{25,26} Trace SO₂ needs to be removed at 0.002 bar with high selectivity in deep desulfurization.^{27,28} Due to the designability and tunability of pore structures as well as high chemical stability, COFs are promising physical absorbents for the separation and recovery of SO₂ but remain to be explored. The isotherm for the adsorption of SO₂ with COF-AB-c was then investigated at 298 K. At 0.002 bar, COF-AB-c uptakes 22 cm³

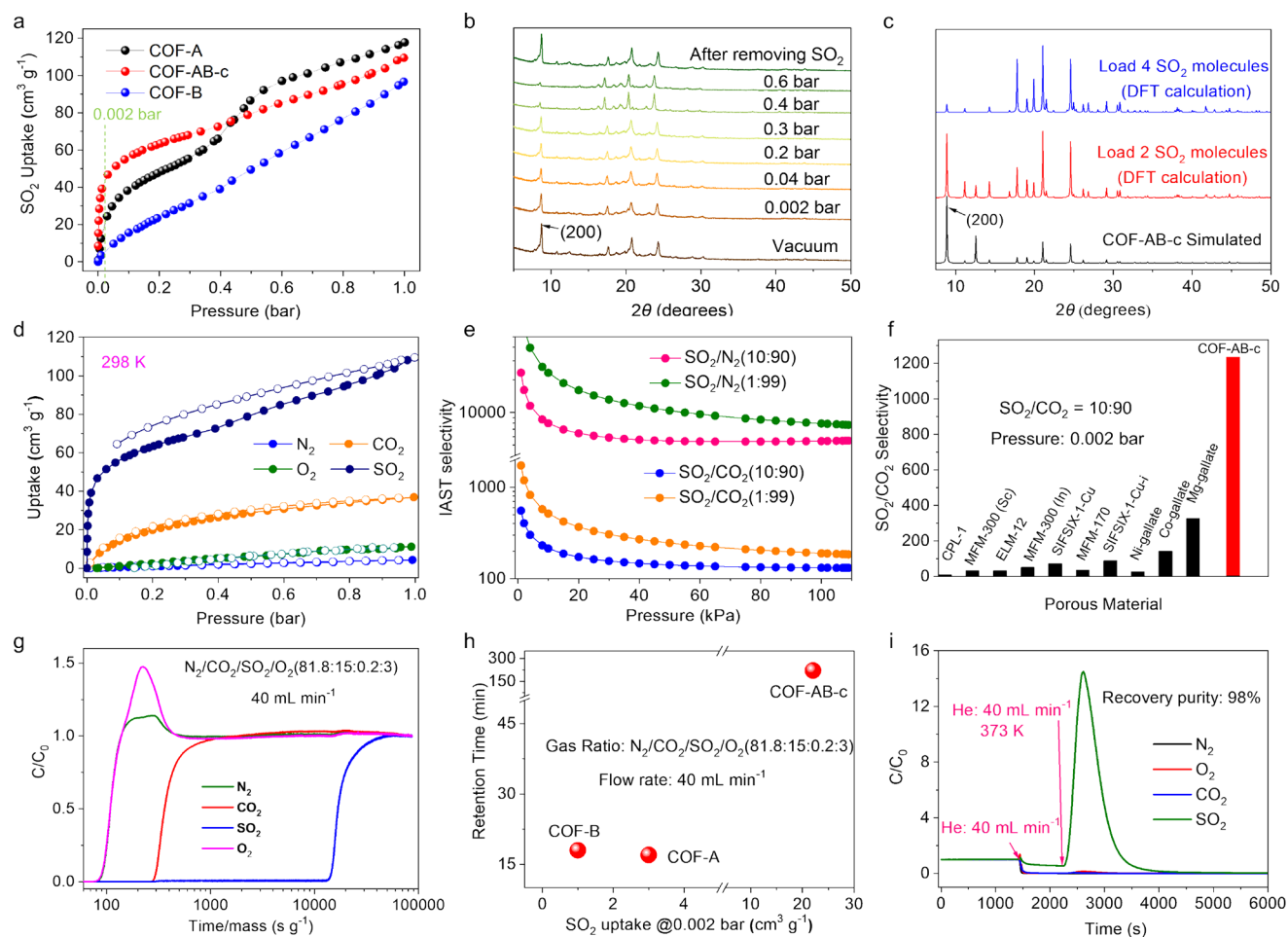


Figure 4. Sulfur dioxide removal from flue gas with COF-AB-c. (a) SO_2 adsorption isotherms of COF-A, COF-AB-c, and COF-B at 298 K. (b) In-situ PXRD patterns under various SO_2 pressures. (c) DFT-calculated powder XRD patterns of COF-AB-c loading two SO_2 and four SO_2 molecules with each unit cell. (d) SO_2 , CO_2 , O_2 , and N_2 adsorption isotherms at 298 K. (e) Ideal solution adsorption theory (IAST) selectivity of SO_2/CO_2 and SO_2/N_2 at 298 K. (f) Ideal adsorption solution theory (IAST) selectivity of SO_2/CO_2 at 298 K for crystalline porous materials. (g) Cycling column breakthrough tests. (h) Comparison of SO_2 adsorption capacity and retention time of COF-A, COF-AB-c, and COF-B. (i) In-situ SO_2 desorption curves.

g^{-1} SO_2 , which is 2200 and 733% of that for COF-B ($1.0 \text{ cm}^3 \text{ g}^{-1}$) and COF-A ($3.0 \text{ cm}^3 \text{ g}^{-1}$), respectively (Figure 4a).

To gain insight into the mechanism behind the enhanced adsorption, the structures of COF-AB-c under different pressures were measured by in-situ powder XRD (Figure 4b). The Bragg peak position of (200) as well as others did not show obvious change, indicating that no lattice parameter change occurred during the adsorption of SO_2 . Meanwhile, the intensity of the (200) peak gradually decreased when the SO_2 pressure increased and a peak at around $14\text{--}15^\circ$ gradually appeared. The reason might be that the adsorbed SO_2 molecules in the pores can cause changes in the phase of powder XRD on different crystal planes of COF-AB-c, resulting in mutual cancellation or superposition and leading to an increase or decrease in the relative intensity of the PXRD diffraction peaks (Figure 4c and see Supplementary Section 8 for a detailed discussion). The main difference between the pore structures of COF-AB-c and its pure-component counterparts is the sequential arrangement of different linkers, which might be the reason for the enhanced adsorption (see Supplementary Section 9 for a detailed discussion). The positive synergy of the uptake capability of COF-AB-c indicated that single-crystal COFs with different components

at controlled ratios might play a vital role in the adjustment of pore structures for specific applications and the discovery of novel material properties.

To investigate the separation performance of SO_2 from flue gas, the adsorption isotherms of CO_2 , N_2 , and O_2 on COF-AB-c were also measured at 298 K. COF-AB-c adsorbed negligible amounts of N_2 and O_2 and exhibited a superior uptake capacity for SO_2 compared to CO_2 (Figure 4d). The selectivity of SO_2/CO_2 was 1230 and 4247 based on ideal adsorption solution theory (IAST) for a mixture of SO_2/CO_2 ratios of 10:90 and 1:99 (Figure 4e), which are typical ratios in flue gas. In contrast, the highest selectivity previously reached by porous crystals was 325 for Mg-gallate with a SO_2/CO_2 ratio of 10:90 at 298 K (Figure 4f).²⁹

To evaluate the feasibility of COF-AB-c for deep desulfurization, a dynamic column breakthrough experiment was performed with a gas mixture of $\text{N}_2/\text{CO}_2/\text{O}_2/\text{SO}_2$ (81.8:15.3:0.2, v/v/v/v) containing 2000 ppm of SO_2 (Figure 4g). COF-AB-c exhibits a desired gas elution order of N_2 and $\text{O}_2 > \text{CO}_2 \gg \text{SO}_2$, manifesting excellent separation performance with N_2 , O_2 , and CO_2 breaking through the fixed bed rapidly while SO_2 elutes after a sufficiently long time with a flow rate of 40 mL min^{-1} . COF-AB-c displays a retention time

of 14037 s g⁻¹ (233.95 min g⁻¹), which is approximately 1300% of that of COF-A (1020 s g⁻¹, 17 min g⁻¹) and COF-B (1080 s g⁻¹, 18 min g⁻¹) at 298 K and 1 bar (Figure 4h). After breakthrough of SO₂, the packed bed was regenerated by heating at 373 K under helium flow for 40 min (Figure 4i). After that, no SO₂ or CO₂ was detected by mass spectrometry in the effluent stream, indicating the complete regeneration of COF-AB-c. According to the desorption curve, the purity of the recovered SO₂ reached 98%, which could be directly used for oxidation to prepare sulfuric acid or for drug synthesis in industry.^{24,30} Six cycles of column breaking through experiments were performed, and SO₂ capture remained unchanged, indicating good durability and recyclability of COF-AB-c (Supplementary Section 9).

CONCLUSIONS

In summary, we showed that linker-A could guide other linkers (linker-B to linker-I) that form either COFs (COF-B, COF-C, COF-D, and COF-E) or amorphous polymers (aCOF-F, aCOF-G, and aCOF-H) with the node to gain single-crystal COFs to form nine types of mixed-component single-crystal COFs with bulk purity and controlled ratios. Single-crystal COFs with up to nine different components were obtained, which are more complex than any known crystals. The lattice structures of these COFs could be systematically tuned by varying the ratios of different components, which are homogeneously distributed on the unit cell scale from the first beginning to the end of the crystallization. Different components in single-crystal COFs lead to complex and diverse pore structures and offer the possibility to discover new properties. As an example, COF-AB-c exhibited 2200 and 733% SO₂ uptake capacity of COF-B and COF-A at 298 K and 0.002 bar, respectively. The selectivity of COF-AB-c for the separation of SO₂/CO₂ ranged from 1230 to 4247 for flue gas based on the ideal adsorbed solution theory, recording porous crystals. Dynamic breakthrough experiments showed that the retention time for SO₂ on COF-AB-c could be enhanced by a factor of 1300% in comparison to that of COF-A and COF-B.

EXPERIMENTAL SECTION

Methods. Synthesis Procedures of Single-Crystal COFs. General procedure A: To grow single-crystal COFs, we dissolved C₁₆-GlyA (251 mg, 0.8 mmol) and sodium hydroxide (32 mg, 0.8 mmol) in 80 mL of water. Then, 4.0 mL of 0.2 mol L⁻¹ of hydrochloric acid was added to neutralize the solution. The solution was stirred for more than 10 min, and then a solution of tetrakis(4-aminophenyl)methane (76 mg, 0.2 mmol) in 0.1 mol L⁻¹ *p*-toluenesulfonic acid (8 mL) was added and stirred well. Then, 0.4 mmol of single or mixed linker (A, B, C, D, E, F, G, H, I) was added and stirred for 5 min at 50 °C. After the completion of the reaction, the product was collected by filtration, washed three times with water and tetrahydrofuran, and dried at ambient temperature for 12 h and 100 °C for 12 h to afford powders of single-crystal COFs.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/jacs.3c13069>.

Materials and methods; synthesis and characterization of multicomponent COFs; continuous rotation electron diffraction (cRED) analysis; powder XRD analysis; solid-state nuclear magnetic resonance spectroscopic analysis; thermogravimetric analysis (TGA); gas absorp-

tion; optical microscopic pictures of synthesis COFs; discussion on adsorption mechanism of sulfur dioxide; and discussion of the effect of sequence on SO₂ adsorption (PDF)

Accession Codes

CCDC 2289788–2289790 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

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Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

We thank the National Natural Science Foundation of China (52061135103 and 51801238) and Fundamental Research Funds for the Central Universities, Sun Yat-sen University (No. 23yxqntd002), for financial support. Structure characterizations were supported by the Instrumental Analysis and Research Center of Sun Yat-sen University and the Institute of Chemistry.

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